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Advancing Saturated Reservoir Evaluation: Gas Cap Identification and Downhole Constant Composition Expansion Experiment in Volcanic Breccia Formations via New-Generation Formation Testing

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Abstract

The Jiamuhe Formation in the Junggar Basin comprises volcanic breccia reservoirs where fluid typing remains petrophysically challenging. Traditional methods (NMR, triple-porosity logs) suggested oil-bearing zones with free water, while DST data misinterpreted gas cap signals as dissolved gas, obscuring gas-oil contacts (GOC). This study integrates new-generation formation testing (NGFT) with advanced petrophysical logging to address these limitations.

This study systematically demonstrates three key advancements: ① A methodological framework integrating spectroscopy logging, 2D nuclear magnetic resonance (NMR) logging, array dielectric measurements, and conventional triple-porosity logs was established for comprehensive reservoir property evaluation and preliminary sweet-spot screening. The integrated analysis enables lithofacies classification and pore structure characterization. Subsequently, NGFT technology was employed to dynamically resolve fluid properties, including composition, viscosity, density, formation volume factor (FVF), and gas-oil ratio (GOR). Calibration of these downhole-measured fluid parameters significantly enhanced petrophysical interpretation accuracy by recalibrating critical parameters (e.g., Archie exponents, saturation models) used in log-based fluid typing workflows. ② The implementation of in-situ constant compositional experiments (CCE) using NGFT tools is presented. Through real-time downhole fluid analysis during controlled depressurization processes, bubble-point pressures were experimentally determined under reservoir conditions. This data-driven approach facilitated the construction of field-specific phase envelope, enabling precise identification of GOC through thermodynamic equilibrium analysis. ③ Through NGFT, we conducted a systematic dynamic evaluation involving step-rate production testing and pressure buildup analysis. This workflow enabled quantitative determination of key reservoir performance parameters: drainage radius, effective permeability, and productivity indices. The acquired dynamic data (flow rates vs. pressure responses) were analyzed using transient flow theory models to establish reservoir deliverability characteristics. These parameters provide critical empirical constraints for production prediction models, effectively reducing uncertainties associated with conventional log-derived permeability estimations.

The NGFT technique successfully identified the reservoir fluid saturated oil with a gas cap. GOC positions were determined through in-situ CCE and density contrast analysis between gas and oil phases. These accurately characterized fluid properties provided critical constraints for optimizing petrophysical interpretation parameters, effectively resolving the long-standing challenges within volcanic breccia reservoirs. Furthermore, dynamic reservoir evaluation via NGFT enabled precise productivity assessment, with calculated productivity showing remarkable consistency with subsequent DST measurements.

NGFT confirms a saturated oil reservoir with gas cap (GOR: 63.5-65000 m³/m³), resolving historical fluid typing controversies. Recalibrated petrophysical parameters improve water saturation calculations and further oil/gas identification. Productivity predictions align with DST results, validating the integrated approach.

Introduction

Conventional petrophysical log evaluation in volcanic reservoirs faces critical challenges due to complex mineralogy (e.g., magnetite, chlorite) distorting porosity and resistivity measurements. Extreme pore heterogeneity—spanning vesicles, dissolution pores, and fractures—causes resistivity misinterpretations: fractures induce "low-resistivity pay" artifacts, while microporosity-bound water masks movable fluids. Ambiguous log responses hinder gas/oil/water differentiation, with gas effects often obscured and heavy oil mimicking clay-bound water. Archie saturation models fail due to absent universal exponents and clay conductivity effects, inflating water saturation estimates. Critically, static logs poorly predict dynamic behavior: permeability estimates are unreliable (fracture-matrix permeability diverges by 3–5 orders), and gas caps are misidentified as dissolved gas, obscuring gas-oil contacts (GOC). These limitations necessitate integrating advanced logs (spectroscopy, 2D NMR, dielectric) with Wireline formation testing (WFT) to resolve fluid ambiguities and bridge the static-dynamic gap.

Traditional wireline formation testing (WFT) tools integrate inlet, pump-out, and fluid analysis modules. Their fluid inlets primarily consist of single-probe configurations or 1-meter-long dual packers. However, in strongly heterogeneous, low-permeability volcanic reservoirs, these tools exhibit low sealing and operational success rates. Therefore, we introduce a new-generation dual-flowline formation tester featuring packers with a 6.4-meter sealing length.

New Generation Formation Testing Technology (NGFT)

The next-generation wireline formation tester (WFT) integrates novel hardware architecture and digital automation to address extreme reservoir challenges. Hardware Architecture: The core innovation is a modular platform centered on a high-efficiency focused radial probe featuring radially arranged guard inlets (44 in²) and sample inlets (64 in²) within a single packer assembly, delivering a 110 in² total flow area for ultra-low permeability access (<0.01 md/cP), Fig. 1a. Another core innovative design is the dual-flowline dual-packer system, which can be flexibly set across 1.82–15-meter formations based on reservoir evaluation requirements. Its dual-flowline configuration maximizes gravitational segregation of wellbore fluids, enabling expedited detection of formation hydrocarbon indications through the upper flowline inlet Fig. 1b.

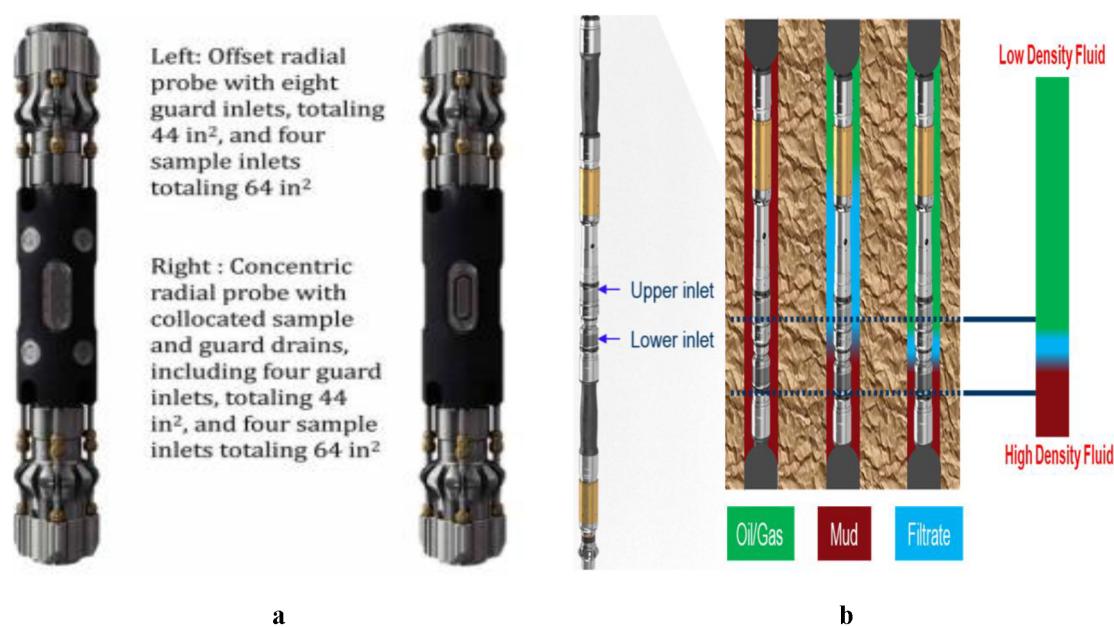


Figure 1—NGFT inlet Module: a. Focused radial Probe b. Dual Packer(1.82~15m sealing length), with illustration of the low/high density fluid segregation

This interfaces with a dual-flowline system connected to two independent, wide-band pump modules ("Flow Manager") capable of 0.1–100 cm³/s flow rates. Contamination-free sampling is enabled by a Sample Manager using motor-driven valves for "glide sampling" (bypassing pumps to preserve trace elements like H₂S) and self-sealing PVT bottles (400cc N₂-compensated or 675cc hydrostatic) with zero-loss transfer (Gama, Villegas 2020, Xiannan, Wang 2023).

The fluid analysis module, FISO (Fluid In-Situ Sample Optimizer) as the new generation DFA(Downhole Fluid Analysis), represents the most accurate and comprehensive downhole fluid analysis (DFA) system currently available. It is capable of distinguishing individual compositions including C1, C2, C3, C4, C5, C6+, and CO₂ with an accuracy within 1.5–3%. Additionally, it can monitor multi-sensor contamination with an accuracy within 2%, and supports high-value limits for gas-oil-water fractions and gas-oil ratio (GOR), reaching up to 50,000 scf/bbl. The module also includes all standard functionalities such as density, viscosity, resistivity, fluorescence, and pH measurement. It features the world's first downhole sensor capable of accurately quantifying fluid resistivity, as well as real-time measurement capabilities with laboratory-level accuracy. For spectral analysis, the module employs a dual-flowline 24-channel spectrometer, enabling full fluid traceability while reducing overall length and weight. Due to its optimized focused sampling and contamination monitoring capabilities, provides decontaminated fluid property measurements, FISO is applicable to all reservoir and fluid types. The architecture of FISO is shown on Fig. 2.

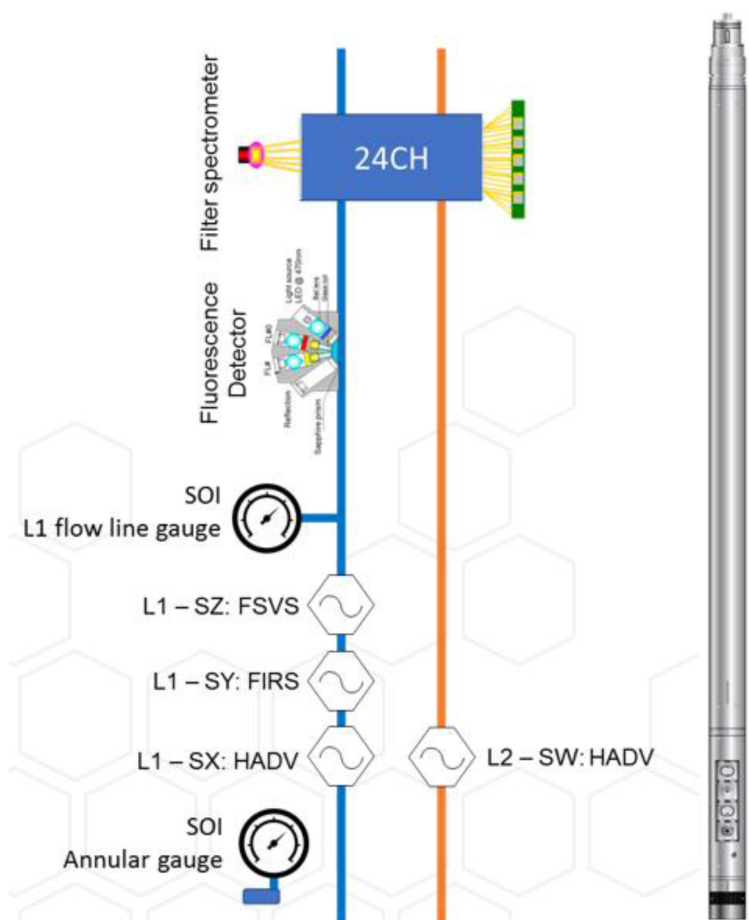


Figure 2—NGFT Downhole Fluid Analysis Module

Formation Testing While Tripping (FTWT)

To perform Deep Transient Testing (DTT), the formation tester must pump substantial volumes of formation hydrocarbons, usually needs pump cubic-meter scale formation fluids. However, as large quantities of hydrocarbons mix into the drilling mud, the annular mud pressure progressively decreases, potentially introducing well control risks. We therefore developed the FTWT technology, with its workflow illustrated in Fig 3.

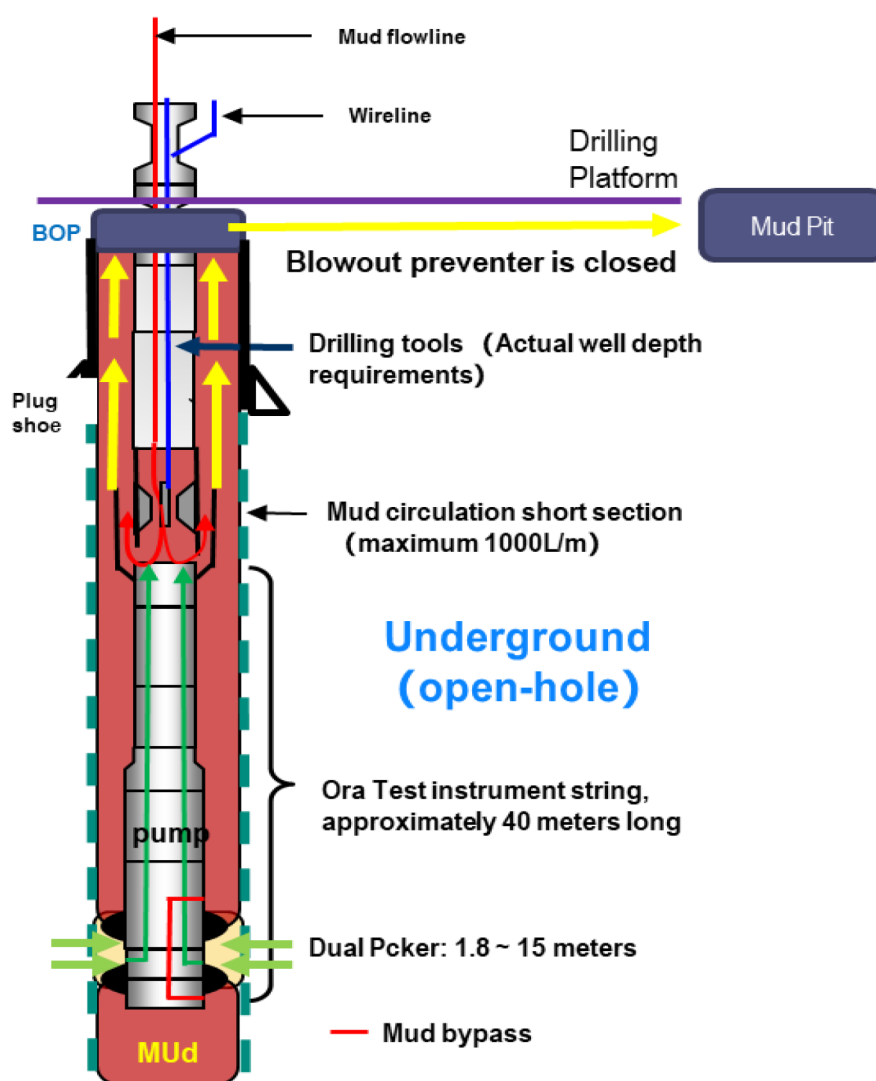


Figure 3—Illustration of Formation Testing While Tripping (FTWT).

The process utilizes adjustable dual packers (1.82–15m span) to isolate the target open hole zone while activating a dedicated mud bypass channel above them. During high-volume formation fluid extraction (e.g., 220L/h) by the ORA tool string (~40m length), rig mud pumps continuously inject fresh mud ($\leq 1,000$ L/min) down the drill pipe-casing annulus. This mud enters the bypass channel to establish an independent circulation loop returning to surface via the choke manifold, which precisely regulates backpressure to compensate for annular pressure loss caused by hydrocarbon withdrawal. Closed blowout preventers (BOPs) and real-time pressure monitoring maintain annular pressure within the safe window ($>$ formation pore pressure to prevent kicks; $<$ fracture pressure to avoid losses). Simultaneously, gravity segregation separates lighter hydrocarbons in the sampling interval, with the upper flowline inlet enabling early hydrocarbon detection. This integrated system decouples formation testing from annular pressure management, enabling safe high-flow-rate deep transient testing while preserving well integrity (Ayan, C. et al., 2017. Jansen Oliveira, 2021).

Geological Background

The triangular Junggar Basin is bounded by the Qingglidi-Kelameili Mountains (NW) and the North Tian Mountain (S). Its basement comprises Precambrian crystalline rock and early to Middle Paleozoic folded rock series (Wang et al, 2024), situated at the Paleo-Kazakhstan, Paleo-Siberia, and Paleo-Tarim plate

intersection, forming part of the Paleo-Kazakhstan plate (Zhang et al, 2006). Shown in Fig. 4, the Zhongguai Uplift, a Late Paleozoic-Mesozoic paleo-high in the basin's northwest, is encircled by hydrocarbon-generating depressions, providing a strategic location for migration. It has experienced deposition from Carboniferous-Permian to Cretaceous-Cenozoic, serving as a persistent paleo-high influencing sediment dispersal (Li et al, 2008). The Early Permian Jiamuhe Formation formed during the closure of the Paleo-Asian Ocean, marking a shift from extensional to compressional regimes (Li et al, 2024). Overlain by the organic-rich Fengcheng Formation source rock, the Jiamuhe Formation is a key target for volcanic reservoir exploration.

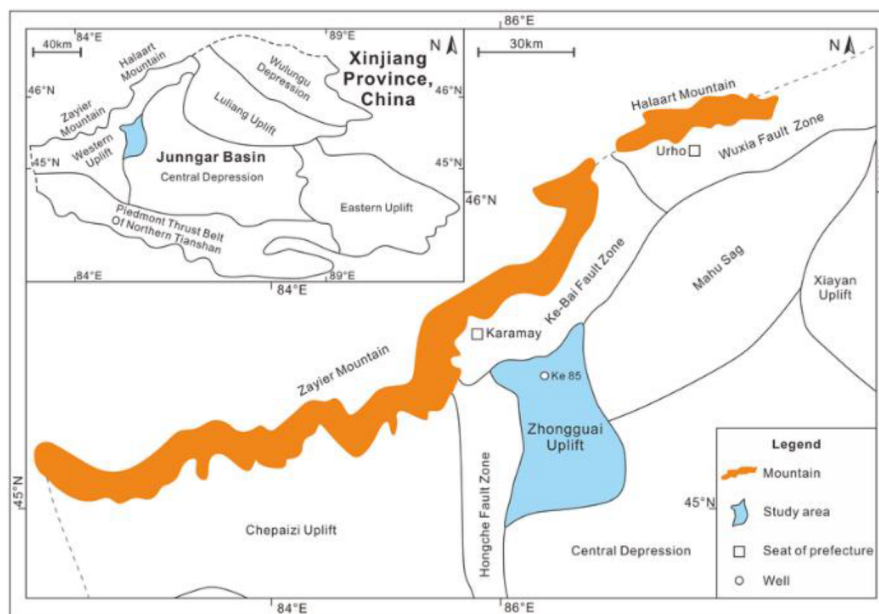


Figure 4—Tectonic map showing the location of Junggar Basin (inset) and Zhongguai Uplift (Shi et al, 2021).

Fluid property is always a headache while performing the formation evaluation. Distinguishing gas, oil and water from the conventional logs is challenging due to extreme heterogeneity. Complex lithologies (volcanic/volcaniclastic rocks), rapid lateral facies changes, and diverse pore systems cause conventional logs to have low contrast, hindering accurate assessment of porosity, permeability, fluid identification, and sweet spot prediction.

Mud Logging and Petrophysical evaluation

Well A, an exploration well in the Zhongguai Block, targeted for the Jiamuhe Formation's volcanic reservoirs, was drilled with full mud logging suite, and Full suite of wireline logging was performed including comprehensive logging, spectroscopy, borehole imaging, T1-T2 based 2D NMR, array dielectric measurement etc.

Reservoir Fluid Estimation based on mud gas is a useful method to resolve the fluid identification challenges (Yang et al, 2023), Yanng developed novel methods to estimate reservoir fluid properties from advanced and standard mud gas (SMG). The ratios include $C1/C2$, $C1/C3$, $C2/C3$, and Bernard ratio ($C1/(C2 + C3)$) (Bernard, 1978; Haworth et al, 1984). Should the corrected mud gas data from a new formation fluid exhibit a gas ratio below the established threshold for oil, it serves as a reliable indicator of an oil-prone formation fluid. As a general guideline, taking the $C1/C2$ ratio as an example, a ratio exceeding 15 typically indicates a gas-prone reservoir fluid, whereas a ratio below 15 signifies an oil-prone reservoir fluid (Pixler, 1969). However, in certain regions, the $C1/C2$ ratio may not distinctly separate different fluid

types. In such cases, there is an overlap in C1/C2 ratios, precluding fluid typing solely based on C1 to C3 compositions (Yang et al., 2021).

As previously stated, a ratio exceeding 15 typically indicates gas-prone reservoir fluid, whereas a ratio below 15 signifies oil-prone reservoir fluid. Across the entire sweet spot area, the C1/C2, C1/C3, and C1/(C2+C3) ratios all exceed 20. Therefore, following this logic, the fluid properties in this well are interpreted as gas.

As shown in Fig. 5, mud gas analysis reveals that within Zone 1, the C1/C2, C1/C3, and C1/(C2+C3) ratios remain stable and are higher than those in Zone 2. In Zone 2, from XX13 m to xx40 m depth, these ratios gradually decrease. This suggests a progressive compositional transition in the formation fluids. Below xx40 m, the fluid properties tend to be consistent.

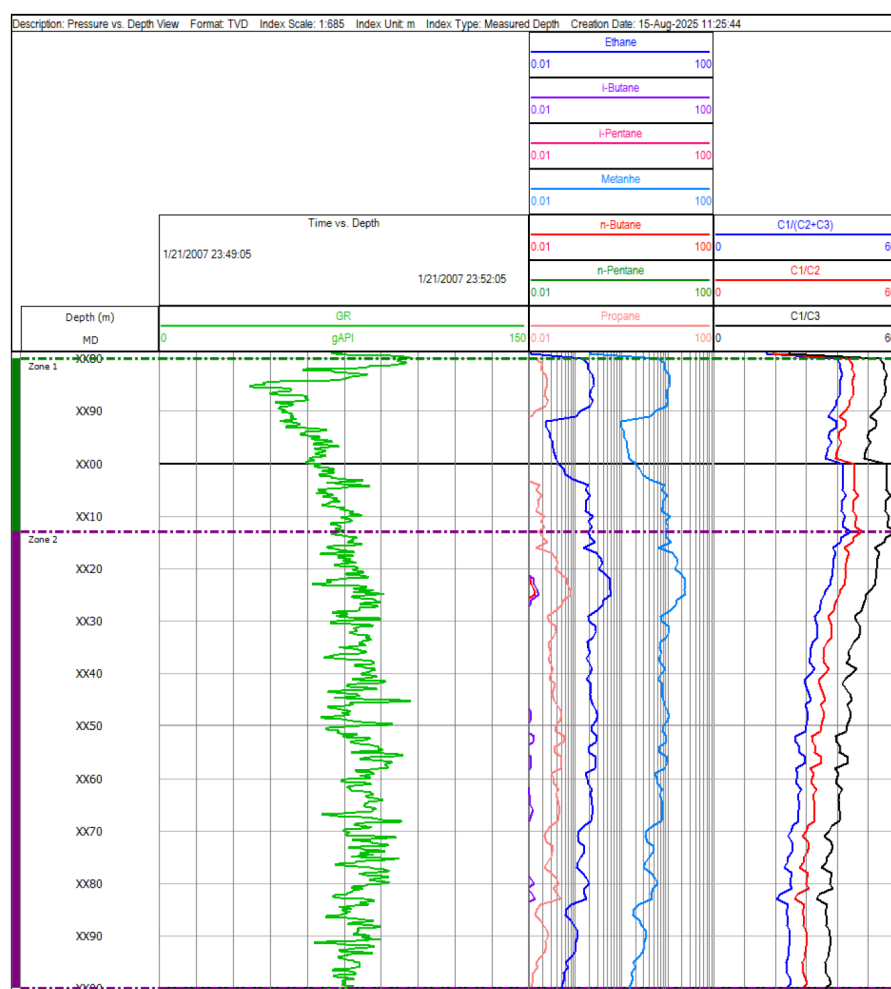


Figure 5—Mud Gas Analysis. Track1 depth; Track2 Gamma ray; Track3 Mud Gas; Track4 mud gas ratios.

Mud gas serves as an effective indicator of variations in formation hydrocarbon composition. However, during early exploration phases, this method proves difficult to quantify, and its results require additional data for verification.

Wireline Logs indicated for the targeted interval; relatively flat response was observed from array laterolog resistivities and density-neutron-compressional logs crossovers. Gamma ray starts a slight pickup since XX10m while the density log reads a little bit lower in the same interval. Spectroscopy showed consistent mineralogy with minor feldspar variations. Borehole images concluded that lithology was volcanic tuffs and tuffaceous sandstone with stable facies and significant breccia, however rare fractures

were observed. Conventional porosity determination was challenging, and resistivity couldn't distinguish hydrocarbon from water.

2D NMR analysis indicated that total porosity of this interval was ranging from 5.5% to 14.5% and Free Fluid Volume (FFV) was between 2% and 7%. T1-T2 mapping confirmed hydrocarbon-dominant free fluid signals alongside substantial bound fluid presence. The measured total water saturation was 55–70%, with effective water saturation of 40–55%. Dielectric measurements reported slightly lower water saturation values (50–65%), supporting the interpretation of hydrocarbon-dominated free pore space with approximately 40% irreducible water saturation (Fig. 6). Confirmed hydrocarbon instead of water was the preliminary fluid during production. Checking all the traditional gas indicators, such as density-neutron crossover, compressional and shear slowness ratios, and Poisson's ratio, were inconclusive. Therefore, while petrophysical log analysis confirms the presence of hydrocarbons, it does not permit definitive identification of the interval as either an oil or gas zone, nor does it allow confirmation of the gas-oil contact (GOC). Under these circumstances, oil was temporarily used as the fluid type for the porosity and saturation computation for the evaluation and computation.

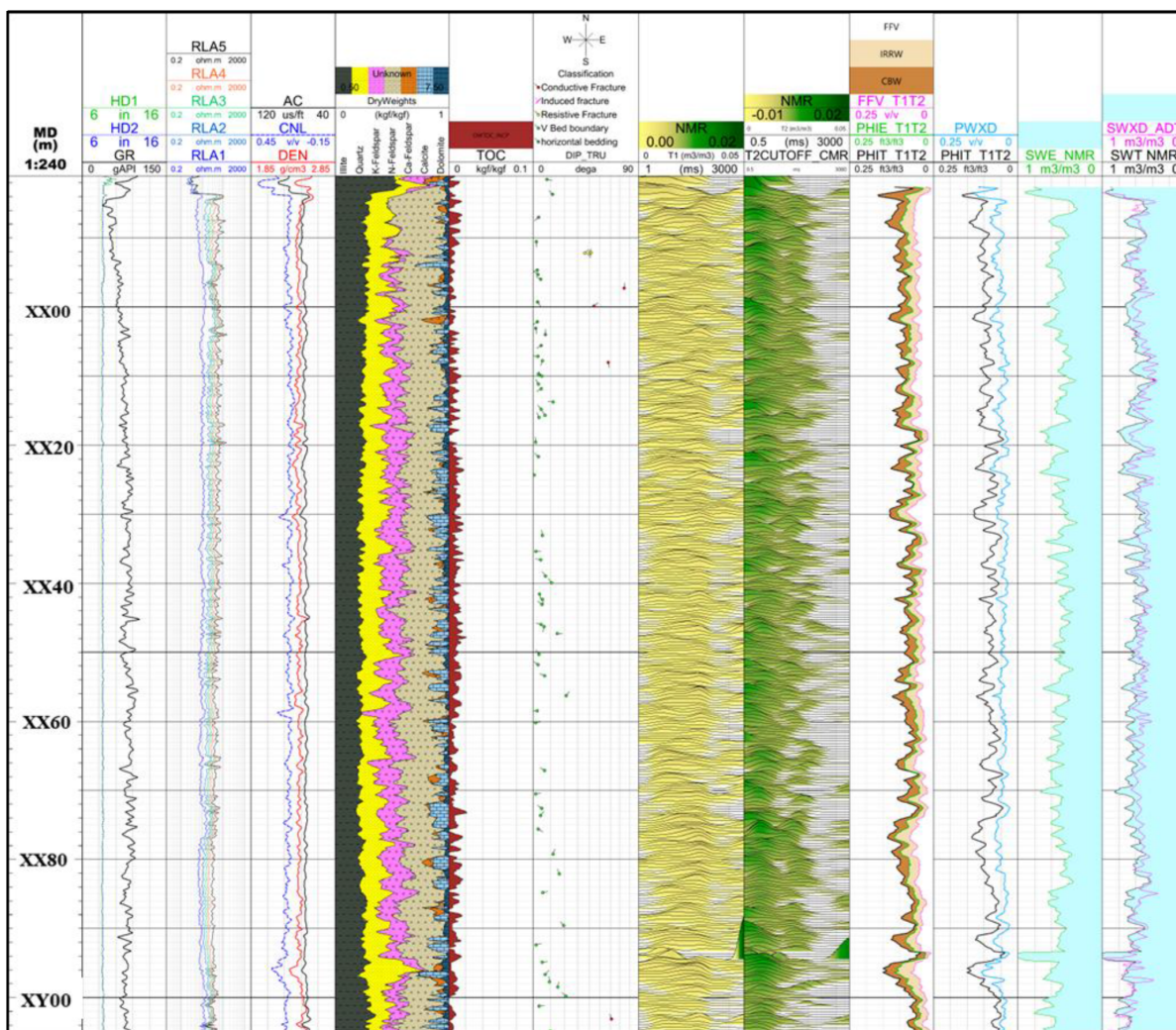


Figure 6—Comprehensive understandings for the well logging suites. Track1 depth; Track2 Caliper and Gamma ray; Track3 Array resistivities; Track 4 Bulk density, neutron and compressional slowness; Track 5 Minerals; Track 6 TOC; Track 7 Dips; Track 8 T1 spectrum, Track 9 T2 spectrum; Track 10 total porosity, effective porosity and free fluid volume; Track 11 total porosity and water porosity from dielectric; Track 12 effective water saturation; Track 13 total water saturation from NMR and dielectric;

Fluid Characterization and GOC confirmation

Fig. 7 is the typical NGFT DTT tool downhole part except drill stem. This tool string (total length 48.25m) integrates a specialized slip joint to address drill stem expansion/contraction during operations. Wellbore temperature changes will cause significant drillstem length variations – the slip joint compensates for up to 1.5m of these movements to prevent tool damage. Other key tool modules include: ①pumping module or Fluid managers (FNFM1/FNFM2) for dual flowlines. ②a real-time downhole fluid analyzer (FISO) measuring optical spectra, viscosity, density, resistivity. ③sample chambers (FSCM, 6×675cc bottles). ④ Dual Inlet and Dual Packer interval isolation module(FCII), 1.82-15m flexibility, adopted 6.4m in this case.

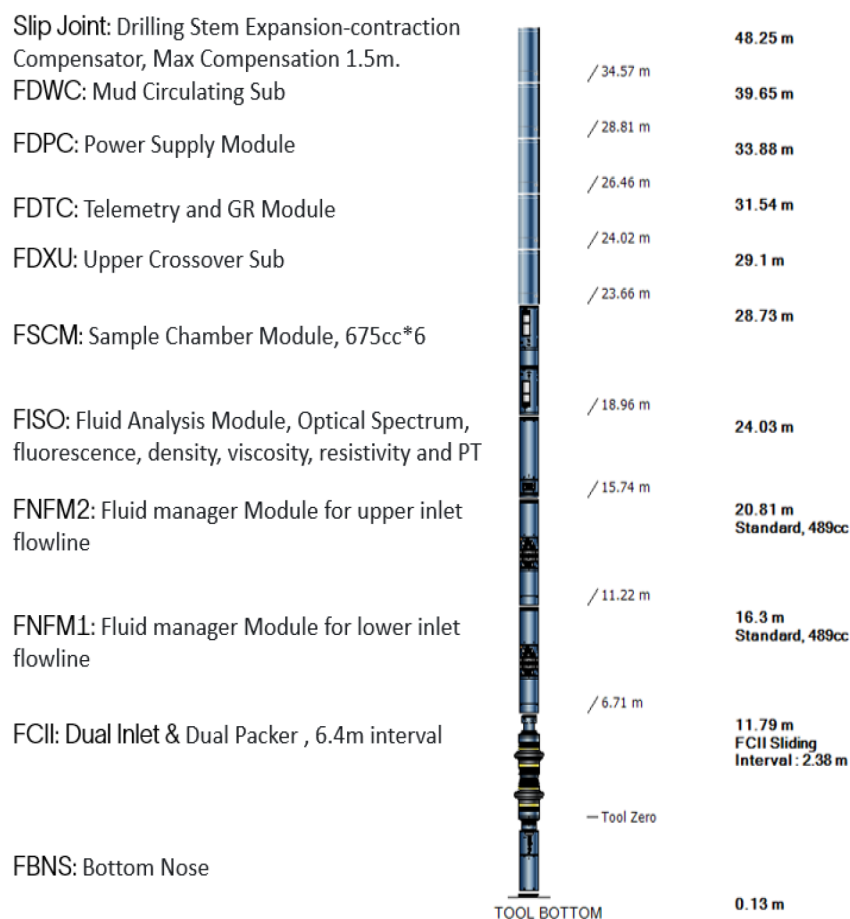


Figure 7—NGFT DTT tool Downhole part without Drill Stem

Through formation testing of the Jiamuhe Formation, distinct fluid properties were identified across three packer-isolated intervals, as Fig. 8 shown.

- A : Interval xx85-xx91.4m: Gas-bearing zone; Fluid volume extracted: 1,691 L; GOR 65084 m³/m³, Composition: C1 86%, C2 5%, C3 4%, C4 2%, C5 1%, C6+ 2%, Fluid Density 0.28g/cm³, Viscosity measurement not reliable.
- B : Interval xx28-xx34.4m: Oil-bearing zone, Fluid volume extracted: 255L; GOR 127m³/m³, Composition: C1 8%, C2 0%, C3 0%, C4 0%, C5 0%, C6+ 92%, Fluid Density 0.75g/cm³, Viscosity 2.5cp.
- C : Interval xx78-xx84.4m: Oil-bearing zone, Fluid volume extracted: 253 L; GOR 63.5m³/m³, Composition: C1 4%, C2 0%, C3 0%, C4 0%, C5 0%, C6+ 96%, Fluid Density 0.77g/cm³, Viscosity 3.6cp.

The fluids exhibit distinct vertical heterogeneity, confirming the reservoir as a saturated oil accumulation with a gas cap, where the GOR (Gas-Oil Ratio) progressively decreases with increasing depth.

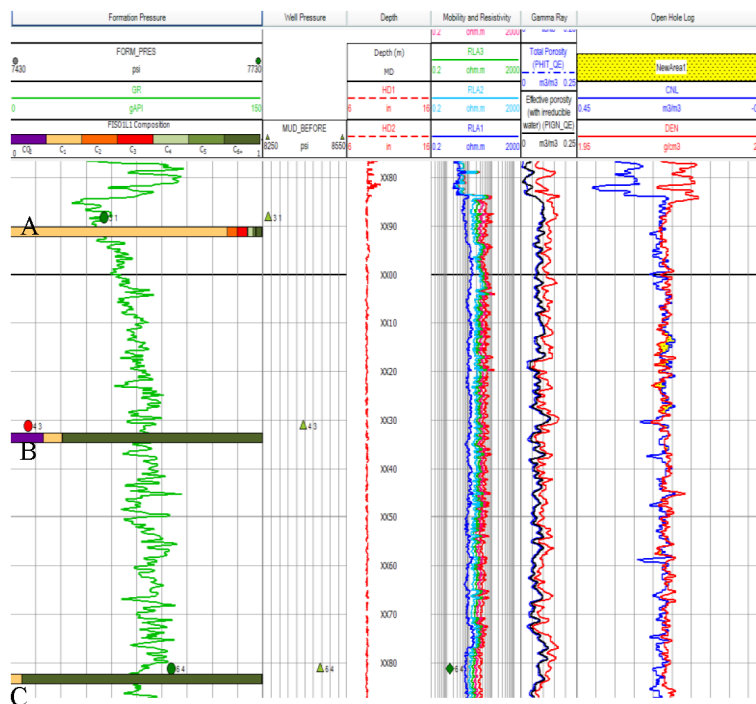


Figure 8—Vertical fluid distribution proved by NGFT

Determining the gas cap thickness is critical for reserve calculation and development planning. This study utilizes downhole density data to analyze the gas-oil contact (GOC). Since the FISO fluid analyzer is downstream of the FNFM module and directly exposed to mud pressure, measured fluid densities must be converted to formation pressure conditions:

- A : Gas zone (xx85-xx91.4m):
Measured density: 0.28g/cm³ @ mud pressure (8,300 psi); Converted density: 0.254g/cm³ @ formation pressure (7541.38psi).
- C : Oil zone (xx78-xx84.4m):
Measured density: 0.77g/cm³ @ mud pressure (8,435 psi); Converted density: 0.72g/cm³@formation pressure (7621.02 psi).

An auxiliary line analysis of pressure versus fluid density confirmed the gas-oil contact at xx11.64 m, as Fig. 9 shown.

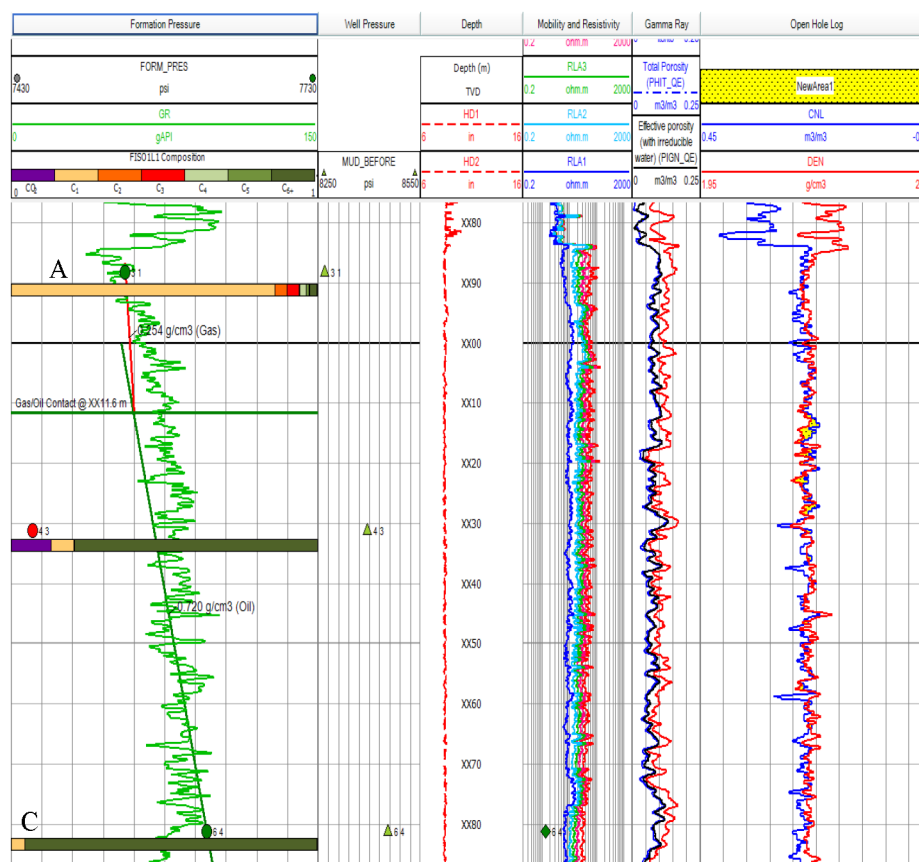


Figure 9—GOC confirmation by fluid density analysis

The above results did not utilize the fluid density or formation pressure at Point B, as reliable formation pressure was not obtained there. However, analysis of the formation fluid variations between Points B and C indicates that density gradually increases with depth, the gas-oil ratio (GOR) decreases, and the heavy hydrocarbon content progressively increases. Therefore, given the oil density of 0.72 g/cc at Point C, it is reasonable to infer an upper limit of 0.68 g/cc for the average fluid density in the oil column. Performing GOC regression analysis suggests the GOC position is at xx05.1m. Consequently, we conclude that the GOC for the entire reservoir lies between xx05.01m and xx11.6m as Fig. 10 shown. Regarding the gas zone, density variations are minimal and will not be further analyzed here.



Figure 10—GOC sensitivity analysis

Calibration of Petrophysical logs interpretation based on NGFT

To provide a clearer understanding for the graphical representation, the GOC results obtained from NGFT were utilized to recalibrate both porosity and hydrocarbon saturation values. Specifically, gas was assigned above xx11.6m, where the bulk density is measured at 0.254g/cm^3 , and oil was designated below this depth, with a bulk density of 0.72g/cm^3 . These parameters were determined using an optimized element analysis approach, allowing for more precise differentiation between the gas and oil zones based on fluid density transitions.

The newly calibrated porosity and corrected water saturation (SUWI_CORR) are displayed in the last two tracks of Fig. 10, which provides a comprehensive visualization of the well logging suite following gas correction. In the recalibrated profiles, porosity is observed to increase by approximately 1–2 porosity units (pu), reflecting a subtle yet meaningful enhancement in the reservoir's storage capacity. Meanwhile, hydrocarbon saturation remains in close agreement with the values documented in the zone, indicating a consistent distribution of hydrocarbons throughout the interval.

This is in line with petrophysical expectations: the lithology and mineralogical composition remain largely unchanged between the upper and lower sections, supporting stable porosity values and suggesting that slightly higher saturations in the upper section are credible. The graphical display in Fig. 10 further illustrates these calibrated trends, showing the clear contrast in porosity and water saturation across the GOC, and highlighting the effectiveness of the gas correction applied to the data.

Additionally, the total organic carbon (TOC) log, included in the composite plot, reveals that TOC values above XX14 are notably lower than those found at greater depths. This trend assists in further distinguishing the gas and oil intervals is visually apparent in the graph, contributing to a more nuanced understanding

of reservoir quality and hydrocarbon potential. However, this is a qualitative indicator and accurate GOC depth cannot be determined.

By integrating the GOC data from NGFT, these petrophysical corrections not only enhance the reliability of the well log interpretation but also provide a robust graphical foundation for subsequent reservoir development and modeling. The comprehensive display in Fig. 11 enables geoscientists and engineers to quickly assess the impacts of gas correction on key reservoir properties, facilitating more informed decision-making for future field operations.

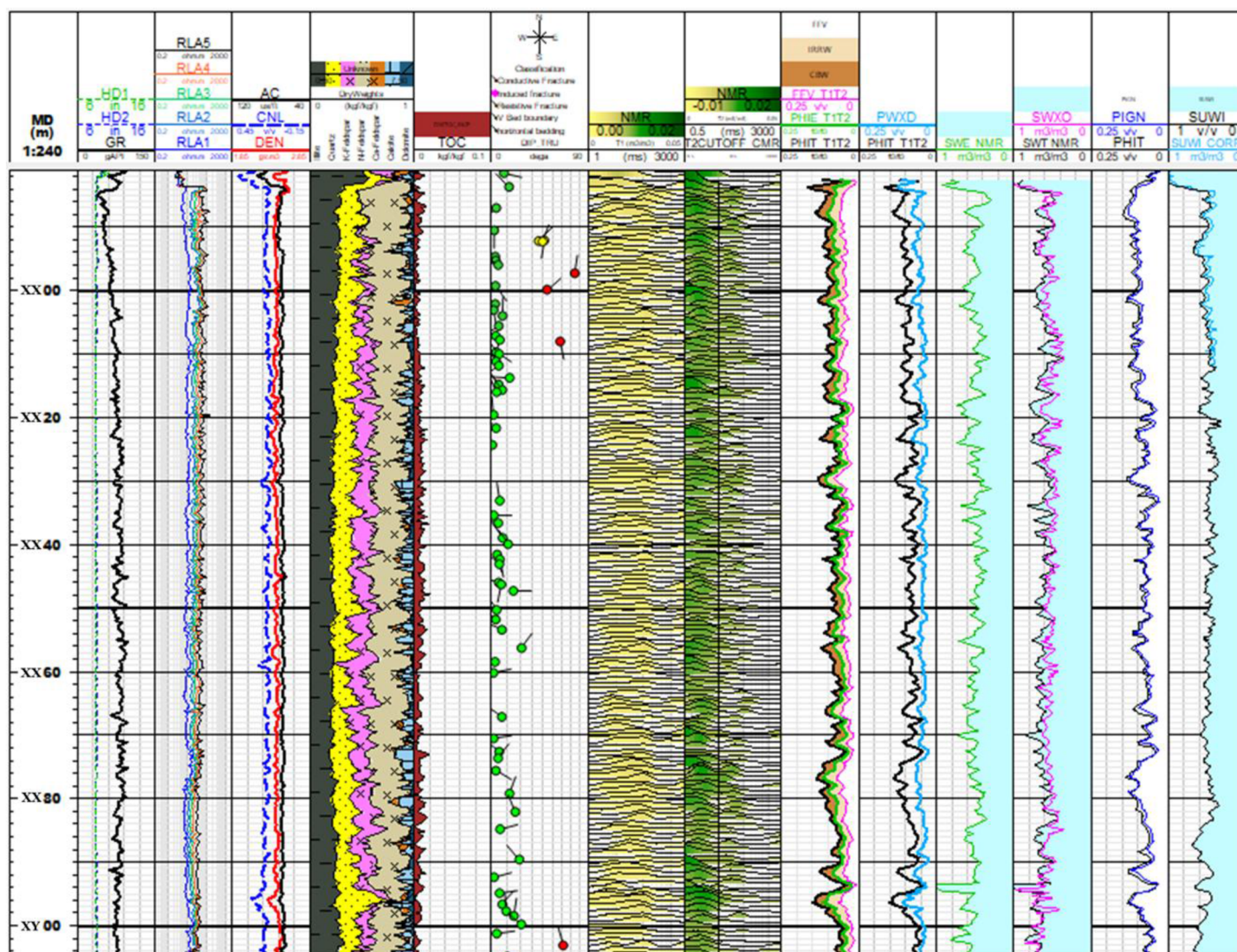


Figure 11—Comprehensive understandings for the well logging suites with gas correction

Downhole Constant Composition Expand Experiment

Utilizing precise flow rate control (0.1-100 cc/s), a downhole CCE was conducted in the oil zone Point B (xx28-xx34.4m) with a minimal pump rate to depressurize the isolated flowline. Dynamic fluid property changes during the CCE were monitored and recorded by the NGFT's downhole fluid analysis module.

Due to the impact of water slugs falling back onto the fluid analysis module's window, noticeable noise appeared in the density-versus-pressure data. This introduced uncertainty into the calculation of FVF (Formation Volume Factor) and compressibility. However, by analyzing the pressure trend, a distinct inflection point was observed at 5500 psi. Consequently, it was determined that the bubble point pressure

at this location is 5500 psi (Fig. 12). The captured 675 cc sample was subjected to CCE testing in the laboratory. As shown in Fig. 13, the results confirmed a bubble point pressure of 5,602.8 psi, which is in strong agreement with the downhole CCE data.



Figure 12—Downhole CCE plot, from upper to lower track:(1) flowline pressure curve, (2) GOR, (3) Density, (4) Optical Density, (5) Optical Density, (6) Composition, (7) Water Fraction

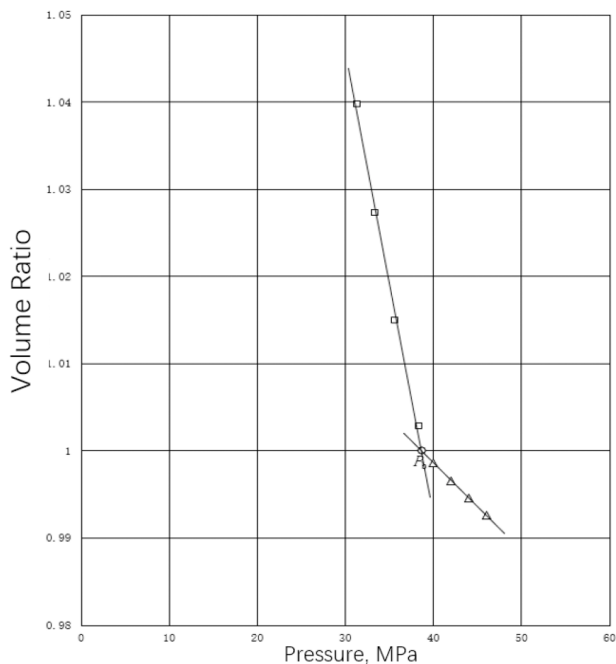


Figure 13—Lab Constant Composition Expanded Experiment, confirmed Pb is 38.64Mpa.

We note that the bubble point pressure at Point B (xx28–xx34.4 m) is 5602.8 psi, which is lower than the gas zone formation pressure of 7541.38 psi at Point A (xx85–xx91.4 m). While this observation initially appears inconsistent with the definition of a saturated oil reservoir, it is explained by the gradual increase in heavier hydrocarbon components with depth. This results in a progressive decrease in local saturation pressure from the top to the base of the oil column, with the highest saturation pressure occurring near the gas-oil contact (GOC). Consequently, there is no fundamental conflict with the concept of an oversaturated reservoir.

Deep Transient Test and Production Prediction

Deep Transient Testing (DTT) fundamentally differs from conventional WFT-based mini-DST through its integration of FTWT technology on next-generation formation testing (NGFT) platforms. DTT operations involve extended pumping phases (tens of hours), cubic-meter-scale fluid extraction (downhole volume), and pressure buildup until definitive Infinite Acting Radial Flow (IARF) signatures are captured. Where seismic data indicate nearby boundaries (e.g., faults, channel edges within tens to hundreds of meters), boundary detection testing may be attempted, potentially extending operations by dozens of hours.

Compared to traditional Drillstem Testing (DST), DTT delivers superior downhole flow rate accuracy and requires minimal wellbore storage (theoretically limited to the annular volume of the isolated interval – e.g., $\sim 0.15 \text{ m}^3$ for a 6.4m packer section in 8.5-in. hole). Consequently, DTT achieves shorter wellbore storage duration and faster progression to radial flow, whereas conventional DST contends with wellbore storage volumes reaching tens of cubic meters and protracted surface production stabilization periods.

The DTT testing process consists of two sequential phases: The fluid characterization phase (1st flow period and subsequent buildup) aims to acquire formation fluid properties during the initial flow, while the buildup assesses near-wellbore damage (skin factor) and cross-validates with the 2nd buildup to evaluate dynamic damage evolution. The productivity evaluation phase (2nd flow period and final buildup) executes a step rate test during the second flow, followed by a buildup to obtain radial flow parameters including skin. As shown in the data, this 13-hour DTT test confirmed dry gas dominance (methane-rich composition), achieving a total pumped volume of 1.6 m^3 with peak flow rates reaching 90 cc/s – significantly exceeding conventional WFT capabilities. The pressure drawdown mobility increased from 2 mD/cp to 4.7 mD/cp , which caused by low viscosity gas increase and damage mitigation. Fig. 14.

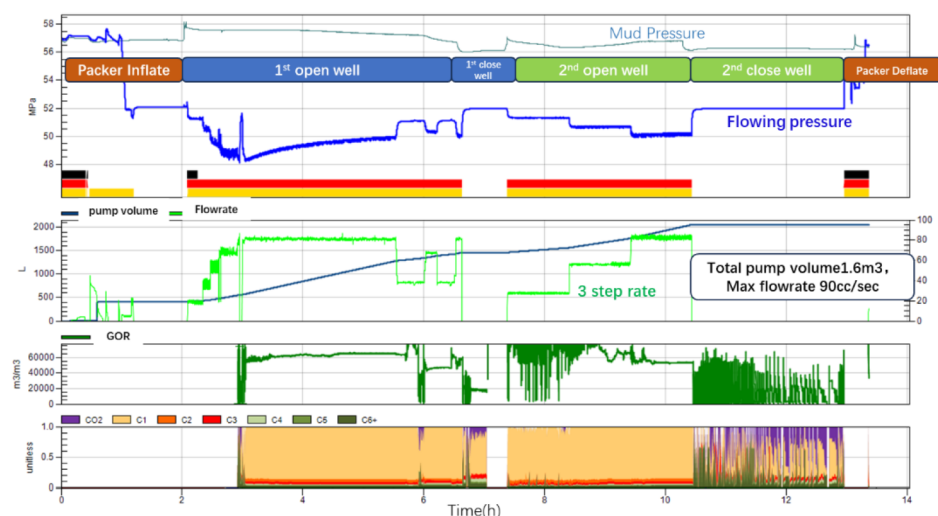


Figure 14—DTT operation fluid and pressure data

Based on Fig.15, the pressure buildup log-log plot demonstrates a distinct Infinite-Acting Radial Flow (IARF) regime. During pressure buildup analysis, the reservoir model adopted was an infinite-acting homogeneous formation, while the wellbore configuration utilized a Vertical Limited Entry model. The following pressure buildup interpretation results were obtained:

- Skin factor $S=40$,
- $Kh/\mu=15.7\text{mD/cp}$
- Investigation Radius= 45m

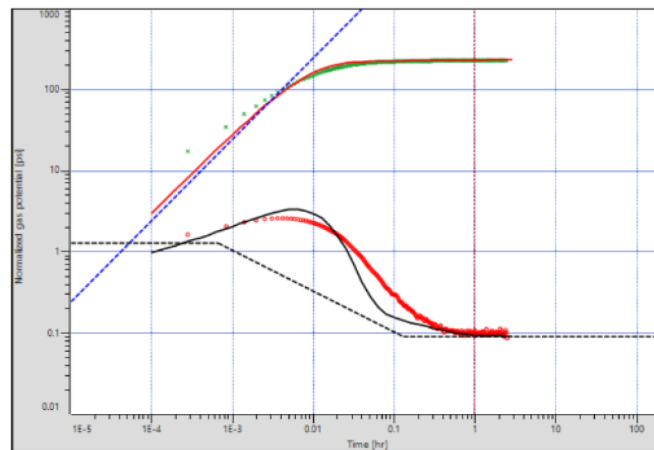


Figure 15—Pressure build up log-log plot

The final history matching results also demonstrate an excellent match between the modeled data and the field-measured pressure data, Fig.16.

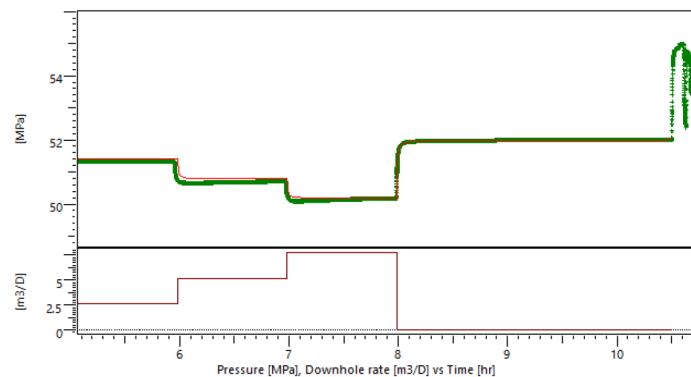


Figure 16—History data match

According to Darcy Law, mobility and Skin factor are critical parameter to determine productivity. Based on the pressure buildup interpretation results, a single-well productivity model was established. The Fig. below presents three inflow performance curves:

- Perforated interval of 6.4m: Production with geometric skin and damage skin. Absolute Open Flow (AOF) = $5.2 \times 10^4 \text{ m}^3/\text{d}$.
- Full gas zone perforated: Production with damage skin. AOF = $5.5 \times 10^4 \text{ m}^3/\text{d}$.
- Full gas zone open: Damage skin reduced from 40 to 0. AOF = $30 \times 10^4 \text{ m}^3/\text{d}$, representing the theoretical maximum productivity without formation damage or stimulation.

Additionally, coordinated productivities under an 8mm choke were simulated as 4.8×10^4 m³/d, 5.0×10^4 m³/d, and 16.1×10^4 m³/d respectively, demonstrating actual deliverability constrained by the 8mm choke, as Fig. 17 shown.

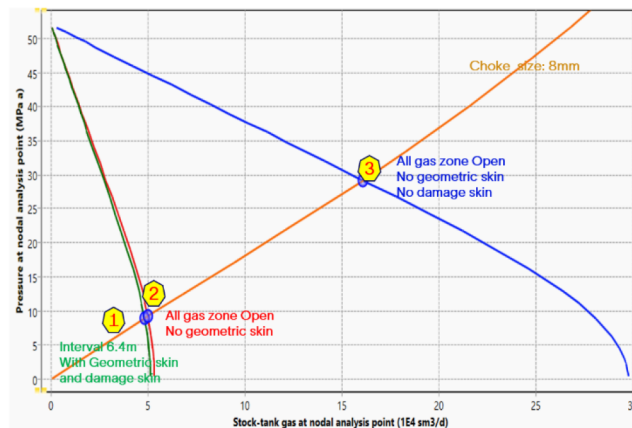


Figure 17—Production performance curve

Perforating a 6.4m interval with geometric skin and damage skin during production replicates the exact conditions of the DTT test process. So theoretically, the predictions of the production forecasting model should match the actual Q-P data from the DTT test.

The Fig. 18 demonstrates that projecting the DTT step-rate test data onto the theoretical IPR curve shows excellent alignment.

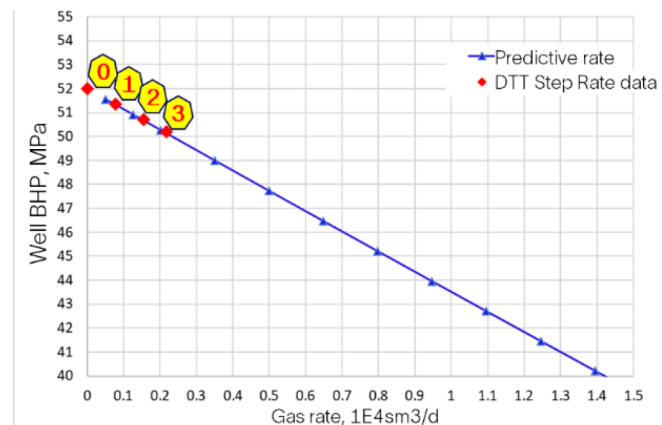


Figure 18—DTT step-rate test data comparison with the theoretical IPR curve

Conclusion

1. Mud gas serves as an effective indicator of variations in formation hydrocarbon composition. However, during early exploration phases, this method proves difficult to quantify, and its results require additional data for verification.
2. The integrated application of NGFT conclusively identified the Jianmhe Formation reservoir as a saturated oil accumulation with a gas cap (GOR range: 63.5 - 65,084 m³/m³), resolving historical uncertainties from conventional logs and DST misinterpretations
3. Gas-Oil Contact (GOC) depth was accurately determined between xx05.1m and xx11.6m using downhole NGFT-derived fluid density measurements converted to formation conditions.

4. Successful downhole Constant Composition Expansion (CCE) experiments performed with NGFT tools experimentally determined bubble-point pressures under reservoir conditions (e.g., 5500 psi at Point B), showing strong agreement with subsequent laboratory PVT analysis (5602.8 psi)
5. Fluid properties and GOC defined by NGFT enabled critical recalibration of petrophysical interpretation parameters (e.g., Archie exponents, saturation models), improving porosity calculations by 1-2 pu and providing more reliable hydrocarbon saturation profiles
6. DTT with NGFT provided robust dynamic reservoir evaluation, quantifying key parameters and enabling precise inflow performance predictions that aligned remarkably well with theoretical models and subsequent DST results.
7. NGFT technology, featuring large-area packers (6.4m seal), dual-flowlines, downhole fluid analysis, and Formation Testing While Tripping (FTWT), effectively addressed the extreme heterogeneity and low permeability of volcanic breccia formations, enabling reliable fluid characterization and dynamic testing previously unattainable

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